# UG (NEP) 4th SEMESTER
# Computer Applications: OOPs with C++
# (CAP422J2)

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No. of Questions: 09 Session: June 2025 Date of Examination: 21-06-2025
Time Allowed: 2:00 Hours Max. Marks: 100 Min. Marks: 36

NOTE: Attempt all questions from Section “A” and “B” and only two questions from Section “C”.

## Section A: Short Answer Type Questions (8 × 2 = 16 Marks)

Answer all questions in a maximum of 20 words each.

i. Define encapsulation?
ii. What is an inline function?
iii. What is a virtual base class?
iv. What is a stream in C++?
v. Define copy constructor?
vi. What are file modes?
vii. What is static member function?
viii. Name any two STL containers?

## Section B: Medium Answer Type Questions (4 × 10 = 40 Marks)

Answer all the four questions with internal choice in a maximum of 150 words each.

2. Explain various types of control structures in C++ with example?

**OR**

Compare procedural and object-oriented programming?

3. Write a note on constructors. How is a copy constructor different from other constructors?

**OR**

Write a program that defines a class Rectangle with length and breadth, includes constructors, and a function to calculate area.

4. Explain how access specifiers affect inheritance?

**OR**

Describe STL. Explain the use of vector and list?

5. Discuss compile-time and run-time polymorphism with examples?

**OR**

Write a short note on file handling functions in C++?

## Section C: Long Answer Type Questions (2 × 22 = 44 Marks)

Answer only two questions in a maximum of 400 words each.

6. Create a program to overload the + operator to add two complex numbers?
7. Write a C++ program to demonstrate class with default and parameterized constructors and destructor?
8. Define virtual function and pure virtual function. How do they support run-time polymorphism in C++?
9. Explain various types of operators available in C++. Illustrate each type with syntax and small examples?